CS604 Midterm Solved McQs Papers By Waqar Sidhu (File 1)

Question No 1

1MB or 1 megabyte is equivalent to

Select Correct Option

- ✓ 1024 bytes
- ✓ 1024^2 bytes
- ✓ 1024^3 bytes
- ✓ **1000000 bytes……(For More Visit VUStudents.pk)**

Question NO 2

A ---------- system collects physically separate, possibly heterogeneous, systems into a single coherent system, providing the user with access to the various resources that the system maintains.

Select Correct Option

- ✓ **Distributed……(For More Visit VUStudents.pk)**
- ✓ Real-time
- ✓ Single user
- ✓ Time-sharing

Question NO 3

---------- has a hierarchical file system structure.

Select Correct Option

- ✓ DOS
- ✓ Windows
- ✓ **UNIX……(For More Visit VUStudents.pk)**
- ✓ None of the given options

Question NO 4

Round-Robin Scheduling is a ---------- scheduling algorithm

Select Correct Option

- ✓ **Preemptive……(For More Visit VUStudents.pk)**
- ✓ Non-preemptive
- ✓ Preemptive or non-preemptive
- ✓ None of the given options

Question NO 5

A piece of code in a cooperating process in which the process may update shared data is called ----------.

Select Correct Option

- ✓ Buffer
- ✓ **Critical section……(For More Visit VUStudents.pk)**
- ✓ Semaphore
- ✓ None of the given options

Question NO 6

The Swap instruction which is the hardware to synchronization problem does not satisfy the _____ condition, hence not consider to be a good solution.

Select Correct Option

- ✓ Progress
- ✓ **Bounded waiting……(For More Visit VUStudents.pk)**
- ✓ Mutual exclusion
- ✓ None of the given

Question NO 7

A _____ is a **piece of code in a cooperating process in which the process may updates shared data (variable, file, database, etc.).**

Select Correct Option

- ✓ **Critical section……(For More Visit VUStudents.pk)**
- ✓ Critical region
- ✓ Monitor
- ✓ All of the given are correct

Question NO 8

Semaphore is a _____ solution to the synchronization problems.

Select Correct Option

- ✓ **Operating system……(For More Visit VUStudents.pk)**
- ✓ Software
- ✓ None of the given
- ✓ Hardware

Question NO 9

In Unix/Linux, by default the standard input file is attached to the _____.

Select Correct Option

- ✓ Mouse
- ✓ **Keyboard……(For More Visit VUStudents.pk)**
- ✓ Light pen
- ✓ Joystick

Question NO 10

The ---------- are used for communication between related or unrelated processes on the same system or unrelated processes on different systems.

Select Correct Option

- ✓ None of the given
- ✓ SSD socket
- ✓ DSD socket
- ✓ **BSD sockets……(For More Visit VUStudents.pk)**

Question NO 11

The files which are automatically opened by the kernel for every process for the process to read its input from and send its output and error messages are called……..,

Select Correct Option

- ✓ None of the given
- ✓ Standard File descriptors
- ✓ File descriptors
- ✓ **Standard files……(For More Visit VUStudents.pk)**

Question NO 12

Bounded Buffer is a buffer of _____ size.

Select Correct Option

- ✓ Variable
- ✓ **Fixed……(For More Visit VUStudents.pk)**
- ✓ Both
- ✓ None of the given

Question NO 13

The goal of single user operating system is

Select Correct Option

- ✓ To maximize utilization of CPU
- ✓ To maximize the user responsiveness
- ✓ **To maximize of peripheral devices……(For More Visit VUStudents.pk)**
- ✓ To maximize throughput

Question NO 14

Which type of operating system guarantees that critical takes must be completed on time?

Select Correct Option

- ✓ Timesharing system
- ✓ Multitasking system
- ✓ Batch system
- ✓ **Realtime system……(For More Visit VUStudents.pk)**

Question NO 15

Which of the following command is used to read data from file named academics?

Select Correct Option

- ✓ con < academics
- ✓ **cat < academics**……(For More Visit VUStudents.pk)
- ✓ socket > academics
- ✓ ran > academics

Question NO 16

_____ command puts a foreground process into background.

Select Correct Option

- ✓ fg
- ✓ **bg**……(For More Visit VUStudents.pk)
- ✓ Ctrl z
- ✓ None of the given

Question NO 17



Given the Gantt chart, which one of the following is average waiting time.

Select Correct Option

- ✓ 35.66
- ✓ **1.33**……(For More Visit VUStudents.pk)
- ✓ 34.66
- ✓ 2.33

Question NO 18

_____ is solution to the problem of indefinite blockage of low-priority processes.

Select Correct Option

- ✓ **Aging**……(For More Visit VUStudents.pk)
- ✓ Starvation
- ✓ Deadlock
- ✓ None of the given

Question NO 19

The part of machine level instruction, which tells the central processor what has to be done, is

Select Correct Option

- ✓ Address
- ✓ Locator
- ✓ Flip-Flop
- ✓ **Operation code**……(For More Visit VUStudents.pk)

Question NO 20
Which of the following is false?

Select Correct Option
- ✓ Two different critical sections may be executed concurrently if they do not use the same shared variables
- ✓ Programs with critical sections can never be used simultaneously by more than one process
- ✓ None of the given
- ✓ A process waiting to enter a critical section currently in use must wait for the process utilizing the critical section to terminate.

(File 2)

Question No 1
When process opens its first file explicitly it will get descriptor number _____.

Select Correct Option
- ✓ 2
- ✓ **3**.....**(For More Visit VUStudents.pk)**
- ✓ 1
- ✓ 4

Question NO 2
Using _____ system call, we can create a new process in Unix.

Select Correct Option
- ✓ **fork**.....**(For More Visit VUStudents.pk)**
- ✓ exec
- ✓ wait
- ✓ exit

Question NO 3
Which one is not the state of the process?

Select Correct Option
- ✓ Waiting
- ✓ Running
- ✓ **Privileged**.....**(For More Visit VUStudents.pk)**
- ✓ New

Question NO 4
To avoid race condition, the number of processes that may be simultaneously inside their critical section is

Select Correct Option
- ✓ 8
- ✓ **1**.....**(For More Visit VUStudents.pk)**
- ✓ 16
- ✓ 0

Question NO 5

_____ loads, verifies, and executes programs that have been translated into Java byte code.

Select Correct Option

- ✓ **Java Virtual Machine……(For More Visit VUStudents.pk)**
- ✓ VMWare
- ✓ VM Solution
- ✓ VM PC

Question NO 6

A time interval when a process uses CPU only is called _____.

Select Correct Option

- ✓ Scheduling
- ✓ I/O burst
- ✓ **CPU burst……(For More Visit VUStudents.pk)**
- ✓ Dispatch latency

Question NO 7

Round Robin schedule usually increase the average waiting time so its performance heavily depends upon _____.

Select Correct Option

- ✓ No. of processes
- ✓ **Time quantum……(For More Visit VUStudents.pk)**
- ✓ Switching
- ✓ No. of queues

Question NO 8

Due to which of the following reason, a parent may terminate the execution of one of its children.

Select Correct Option

- ✓ **Use more resources than allocated……(For More Visit VUStudents.pk)**
- ✓ Busy other than assigned task
- ✓ Start abnormal behavior
- ✓ Many jobs in a system

Question NO 9

Which of the following tool is used for inter process communication between related or unrelated processes on the same system or unrelate processes on different system over the network?

Select Correct Option

- ✓ Pipe
- ✓ **BSD socket……(For More Visit VUStudents.pk)**
- ✓ Name pipe (FIFO)
- ✓ Message Queue

Question NO 10

The files which are automatically opened by the kernel for every process for the process to read its input from and send its output and error messages are called……..,

Select Correct Option

- ✓ None of the given
- ✓ Standard File descriptors
- ✓ File descriptors
- ✓ **Standard files……(For More Visit VUStudents.pk)**

Question NO 11

In Unix/Linux client server process communication, server process runs in the background by terminating its command line with ----------.

Select Correct Option

- ✓ **&……(For More Visit VUStudents.pk)**
- ✓ @
- ✓ %
- ✓ $

Question NO 12

Each thread has its own ----------.

Select Correct Option

- ✓ Process ID (PID)
- ✓ Address Space
- ✓ **Thread ID (TID) ……(For More Visit VUStudents.pk)**
- ✓ Open file descriptors

Question NO 13

Preemptive scheme of Short job first scheduling is know as ---------.

Select Correct Option

- ✓ First Come First Serve Scheduling
- ✓ **Shortest-Remaining-Time-First……(For More Visit VUStudents.pk)**
- ✓ Priority Scheduling
- ✓ Round-Robin Scheduling

Question NO 14

A process terminates by calling the ---------- system call.

Select Correct Option

- ✓ wait
- ✓ fork
- ✓ exec
- ✓ **exit……(For More Visit VUStudents.pk)**

Question NO 15

The -------- system call is used to open or create a file.

Select Correct Option

- ✓ **open()……(For More Visit VUStudents.pk)**
- ✓ read()
- ✓ write()
- ✓ close()

Question NO 16

The syntax for input redirection is

Select Correct Option

- ✓ **command < input-file……(For More Visit VUStudents.pk)**
- ✓ command > input-file
- ✓ command >= input-file
- ✓ command =< input-file

Question NO 17

The syntax for output redirection is

Select Correct Option

- ✓ command < output-file
- ✓ command > output-file
- ✓ **command >= output-file……(For More Visit VUStudents.pk)**
- ✓ command =< output-file

Question NO 18

_____ algorithm is the optimal scheduling algorithm among all the non-preemptive scheduling algorithms.

Select Correct Option

- ✓ **Shortest Job First scheduling……(For More Visit VUStudents.pk)**
- ✓ First Come First Serve
- ✓ Priority Scheduling
- ✓ Round Robin Scheduling

Question NO 19

Process synchronization is a process of helping processes to _____.

Select Correct Option

- ✓ Update the shared data simultaneously
- ✓ **Update the shared data and resources in an orderly fashion**
- ✓ Not to update the shared data or memory
- ✓ Access data and memory at first come first serve basis

Question NO 20

A parent process calling _____ system call will be suspended until any immediate child terminates.

Select Correct Option

- ✓ **wait……(For More Visit VUStudents.pk)**
- ✓ fork
- ✓ exit
- ✓ exec

(File 3)

Question No 1

Main memory is a large array of --------- called memory locations ranging in size from hundreds of thousands of billions.

Select Correct Option

- ✓ Interrupts
- ✓ Registers
- ✓ Digits
- ✓ **Words……(For More Visit VUStudents.pk)**

Question NO 2

In ---------- model, there is a kernel thread for every user thread.

Select Correct Option

- ✓ Many-to-Many
- ✓ None of the given options
- ✓ **One-to-One……(For More Visit VUStudents.pk)**
- ✓ Many-to-One

Question NO 3

A time interval when a process uses CPU only is called

Select Correct Option

- ✓ **CPU burst……(For More Visit VUStudents.pk)**
- ✓ IO burst
- ✓ Scheduling
- ✓ None of the given options

Question NO 4

The part of the program where the shared memory is accessed is called ---------.

Select Correct Option

- ✓ **Buffer……(For More Visit VUStudents.pk)**
- ✓ Critical section
- ✓ Semaphore
- ✓ None of the given options

Question NO 5

Utilities used for system administration (halt, ifconfig, fdisk, etc.) are stored in _____ directory.

Select Correct Option

- ✓ /dev
- ✓ /boot
- ✓ **/sbin……(For More Visit VUStudents.pk)**
- ✓ /lib

Question NO 6

The Swap instruction which is the hardware to synchronization problem does not satisfy the _____ condition, hence not consider to be a good solution.

Select Correct Option

- ✓ Progress
- ✓ **Bounded waiting……(For More Visit VUStudents.pk)**
- ✓ Mutual exclusion
- ✓ None of the given

Question NO 7

All ---------- within a process share the same address space

Select Correct Option

- ✓ **Threads……(For More Visit VUStudents.pk)**
- ✓ Processes
- ✓ CPU burst
- ✓ IO Burst

Question NO 8

In a Multilevel Queue, the foreground queue has -------- scheduling algorithm and background queue has ---------- scheduling algorithm.

Select Correct Option

- ✓ First Come First Serve, Round-Robin
- ✓ **Round-Robin, First Come First Serve……(For More Visit VUStudents.pk)**Round-Robin, Round-Robin
- ✓ First Come First Serve, First Come First Serve

Question NO 9

Priority of a process is ---------- to the priority number of the process.

Select Correct Option

- ✓ Directly proportional
- ✓ Equal
- ✓ Inversely proportional
- ✓ **Greater……(For More Visit VUStudents.pk)**

Question NO 10

The ……. View is that it is a program that acts as an intermediary between a user of a computer and the computer hardware, and makes the computer system convenient to use.

Select Correct Option

- ✓ **Top-down……(For More Visit VUStudents.pk)**
- ✓ Bottom-up
- ✓ Alternate
- ✓ None of the above

Question NO 11

_____ commands in Linux helps to create a new directory.

Select Correct Option

- ✓ ls
- ✓ cp
- ✓ mv
- ✓ **mkdir……(For More Visit VUStudents.pk)**

Question NO 12

_____ scheduler selects the process from the job pool and put them in main memory.

Select Correct Option

- ✓ **Long term……(For More Visit VUStudents.pk)**
- ✓ Short term
- ✓ Medium term
- ✓ Swapper

Question NO 13

---------- are used for communication between unrelated processes on a system.

Select Correct Option

- ✓ Pipes
- ✓ Name pipe (FIFO)
- ✓ **BSD Sockets……(For More Visit VUStudents.pk)**
- ✓ None of the given options

Question NO 14

The syntax for output redirection is

Select Correct Option

- ✓ command < output-file
- ✓ command > output-file
- ✓ **command >= output-file……(For More Visit VUStudents.pk)**
- ✓ command =< output-file

Question NO 15

The shared buffer can be implemented as a circular array with …… logical pointers.

Select Correct Option

- ✓ Five
- ✓ **Two……(For More Visit VUStudents.pk)**
- ✓ Three
- ✓ Four

Question NO 16

Limit register contains

Select Correct Option

- ✓ **The size of the process……(For More Visit VUStudents.pk)**
- ✓ Starting address of the process
- ✓ Address of the next instruction
- ✓ Operand to be fetched

Question NO 17

Which of the following command is used to read data from file named academics?

Select Correct Option

- ✓ con < academics
- ✓ **cat < academics……(For More Visit VUStudents.pk)**
- ✓ socket > academics
- ✓ ran > academics

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_____ command puts a foreground process into background.

Select Correct Option

- ✓ fg
- ✓ **bg……(For More Visit VUStudents.pk)**
- ✓ Ctrl z
- ✓ None of the given

Question NO 19

_____ is solution to the problem of indefinite blockage of low-priority processes.

Select Correct Option

- ✓ **Aging……(For More Visit VUStudents.pk)**
- ✓ Starvation
- ✓ Deadlock
- ✓ None of the given

Question NO 20

What is the requirement for solution to critical section problem?

Select Correct Option

- ✓ Mutual Exclusion
- ✓ Progress
- ✓ Bounded Waiting
- ✓ **All of the given……(For More Visit VUStudents.pk)**

(File 4)

Question No 1

_____ commands in Linux is used to copy file.

Select Correct Option

- ✓ ls
- ✓ **cp……(For More Visit VUStudents.pk)**
- ✓ mv
- ✓ mkdir

Question NO 2

Inter process communication (IPC) allows process to communicate and to synchronize their actions without sharing the same _____.

Select Correct Option

- ✓ **Address space……(For More Visit VUStudents.pk)**
- ✓ Messages
- ✓ Pipes
- ✓ Message Queue

Question NO 3

Pipe system call is used to _____ a pipe.

Select Correct Option

- ✓ Destroy
- ✓ **Create……(For More Visit VUStudents.pk)**
- ✓ Modify
- ✓ Clear

Question NO 4

Select Correct Option

- ✓ con < academics
- ✓ **cat < academics……(For More Visit VUStudents.pk)**
- ✓ socket > academics
- ✓ ran > academics

Question NO 5

In a multithreaded model, Which one of the following provides true concurrency?

Select Correct Option

- ✓ Many-to-one
- ✓ **One-to-one**……(For More Visit VUStudents.pk)
- ✓ Many-to-many
- ✓ One-to-many

Question NO 6

In _____ model, there is a kernel thread for every user thread.

Select Correct Option

- ✓ **One-to-one**……(For More Visit VUStudents.pk)
- ✓ Many-to-one
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Question NO 7

_____ loads, verifies, and executes programs that have been translated into Java byte code.

Select Correct Option

- ✓ **Java Virtual Machine**……(For More Visit VUStudents.pk)
- ✓ VMWare
- ✓ VM Solution
- ✓ VM PC

Question NO 8

_____ is the time taken for the dispatcher to stop one process and start another process.

Select Correct Option

- ✓ **Dispatch latency**……(For More Visit VUStudents.pk)
- ✓ Scheduling
- ✓ Context switching
- ✓ Preemption

Question NO 9

A time interval when a process uses CPU only is called _____.

Select Correct Option

- ✓ Scheduling
- ✓ I/O burst
- ✓ **CPU burst**……(For More Visit VUStudents.pk)
- ✓ Dispatch latency

Question NO 10

A time interval when a process uses I/O devices only is called _____.

Select Correct Option

- ✓ Scheduling
- ✓ **I/O burst……(For More Visit VUStudents.pk)**
- ✓ CPU burst
- ✓ Dispatch latency

Question NO 11

While two processes in execution, only one can be in critical section at a time, this requirement is named as _____.

Select Correct Option

- ✓ Race condition
- ✓ **Mutual exclusion……(For More Visit VUStudents.pk)**
- ✓ Progress
- ✓ Bounded waiting

Question NO 12

The file descriptor for Standard Output (stdout) is ----------.

Select Correct Option

- ✓ 0
- ✓ **1……(For More Visit VUStudents.pk)**
- ✓ 2
- ✓ 3

Question NO 13

The file descriptor for Standard error (stderr) is ----------.

Select Correct Option

- ✓ 0
- ✓ 1
- ✓ **2……(For More Visit VUStudents.pk)**
- ✓ 3

Question NO 14

---------- are used by shell commands to pass data from one shell pipeline to another, without creating temporary files.

Select Correct Option

- ✓ Pipes
- ✓ Message queues
- ✓ BSD sockets
- ✓ **FIFOs……(For More Visit VUStudents.pk)**

Question NO 15

Preemptive ----------scheduling is sometimes called shortest-remaining-time-first scheduling.

Select Correct Option

- ✓ First-Come-First-Served (FCFS)
- ✓ Round-Robin
- ✓ **Sorted Job First (SJF) ……(For More Visit VUStudents.pk)**
- ✓ Priority

Question NO 16

In a Multilevel Queue, the foreground queue has -------- scheduling algorithm and background queue has ---------- scheduling algorithm.

Select Correct Option

- ✓ First Come First Serve, Round-Robin
- ✓ **Round-Robin, First Come First Serve……(For More Visit VUStudents.pk)**Round-Robin, Round-Robin
- ✓ First Come First Serve, First Come First Serve

Question NO 17

A process terminates by calling the ---------- system call.

Select Correct Option

- ✓ wait
- ✓ fork
- ✓ exec
- ✓ **exit……(For More Visit VUStudents.pk)**

Question NO 18

The syntax for input redirection is

Select Correct Option

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A _____ is a piece of code in a cooperating process in which the process may updates shared data (variable, file, database, etc.).

Select Correct Option

- ✓ **Critical section……(For More Visit VUStudents.pk)**
- ✓ Critical region
- ✓ Monitor
- ✓ Semaphore

Question NO 20

A parent process calling _____ system call will be suspended until any immediate child terminates.

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- ✓ **wait……(For More Visit VUStudents.pk)**
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The END

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